

The Θ – Ψ Framework: A Geometric Solution to Large-Scale Structure Anomalies (Paper I)

TERRY SCHERMERHORN¹

¹*Independent Researcher, Brooksville, Florida*

ABSTRACT

Recent spectroscopic surveys (DESI Y1, Euclid Early Release) reveal that the largest cosmic voids ($R > 150$ Mpc) are significantly rounder ($e = 0.11 \pm 0.04$) than Λ CDM simulations predict ($e = 0.35$ – 0.55) — a 4 – 5σ discrepancy with no conventional explanation. We propose that this “roundness reversal,” along with flat rotation curves, cluster mass excesses, the CMB Cold Spot, and the Hubble tension, arises from a single geometric source: a static tensor Θ projected from frozen 5D bulk curvature onto our 4D brane.

This bulk mode is wavelength-limited ($\lambda \approx 120$ Mpc), near-monochromatic, and reconstructible through joint analysis of weak lensing, void morphology, and redshift-space distortions. When superposed with ordinary baryonic curvature Ψ , the resulting Θ – Ψ interference produces all observed large-scale structure features above ~ 10 Mpc without invoking non-baryonic matter. The framework yields 11 falsifiable predictions testable by Euclid, Rubin/LSST, DESI, PTA networks, and CMB-S4 between 2026–2035. A single decisive failure suffices to rule out the pure-geometric variant; survival of all tests would strongly constrain particle dark matter on > 50 Mpc scales.

The explicit functional form $\Theta(r)$ and its brane projection are derived in Appendix A using standard Shiromizu–Maeda–Sasaki formalism with a frozen soft-wall bulk mode. No new fields or free parameters beyond the observed ~ 120 Mpc scale are introduced. This work (Paper 1) outlines observational predictions. Theoretical derivations and bulk-mode stabilization are reserved for Paper 2.

Keywords: cosmology: large-scale structure — dark energy — modified gravity — cosmic voids

1. INTRODUCTION

1.1. Contemporary Cosmology and Λ CDM Successes

Contemporary cosmology describes the Universe to high precision using general relativity with ordinary baryons, radiation, photons, and neutrinos, supplemented by a cosmological constant Λ . (Planck Collaboration 2018) The resulting Λ CDM paradigm successfully accounts for the cosmic microwave background (CMB), Big-Bang nucleosynthesis, and the late-time accelerated expansion. (A. G. Riess et al. 2022)

1.2. Persistent Tensions in Λ CDM

Despite its phenomenological success, the particle-dark-matter paradigm has faced persistent challenges: no confirmed direct-detection signal after four decades, growing small-scale problems (core-cusp, missing satellites, too-big-to-fail), and an increasingly crowded landscape of competing candidates. (J. S. Bullock & M.

Boylan-Kolchin 2017; A. Del Popolo et al. 2017) Simultaneously, the observed emptiness, size distribution, and near-spherical shape of the largest cosmic voids ($R \gtrsim 150$ Mpc) remain difficult to reconcile with the expected tidal stretching in a Gaussian random field seeded by inflation. (S. Nadathur et al. 2021; S. Contarini et al. 2024) These tensions, while individually subdominant, collectively motivate exploration of whether some or all “dark-matter” phenomena could arise from geometry rather than new particles.

1.3. The Θ – Ψ Framework

We explore a framework in which a static, wavelength-limited tensor Θ — interpreted as the projection of frozen 5D bulk curvature onto our 4D brane — acts as an effective stress-energy source on scales larger than ~ 10 Mpc. (T. Shiromizu et al. 2000) The geometric term Θ is not introduced to fit existing data retroactively; instead, recent void catalogs already exhibit the morphology it predicts, most notably the observed 4 – 5σ roundness reversal (Section 2). When superposed with ordinary baryonic curvature Ψ , the resulting Θ – Ψ inter-

ference pattern naturally sculpts sharp filaments, near-spherical supervoids, and radially aligned void-centric structures. We refer interchangeably to the frozen bulk mode Θ , its projected curvature, and its associated gradient $\nabla\Theta$, all representing the same geometric structure viewed at different levels.

1.4. Scale-Dependent Effects

The same geometric term modifies the effective gravitational potential on smaller scales, producing elevated dynamical masses in galaxy clusters and yielding flat rotation curves through a scale-dependent correction that mimics a $1/r$ tail on kiloparsec to megaparsec scales. The characteristic ~ 120 Mpc wavelength emerges from the bulk geometry itself rather than being tuned to observations: it is derived from first principles (Appendix A.1–A.2) using only the standard warped extra-dimension hierarchy scale, and all spatial predictions (void radius $\approx \lambda/2$, beat period in $P(k)$, Cold Spot angular size) follow directly from geometry.

1.5. Falsifiability and Testable Predictions

The framework is deliberately constructed to be falsifiable on a short timescale (2026–2035) using only approved, currently operating, or firmly funded facilities: Euclid, the Rubin Observatory (LSST), DESI, the CMB-S4 experiment, pulsar-timing arrays, and the upcoming LISA mission. A single decisive failure among the eleven independent tests presented below is sufficient to rule out the pure-geometric variant outright; survival of all major tests would constitute strong evidence that at least the largest-scale dark-matter phenomena are geometric in origin. Intermediate outcomes naturally accommodate hybrid interpretations in which particle dark matter dominates only below ~ 10 Mpc. This orthogonality limits speculation. Because Θ represents a frozen bulk geometric projection rather than a dynamical 4D field, it is absent from standard N -body simulations regardless of resolution or baryonic subgrid physics.

2. SUPERVOID MORPHOLOGY: AN EXISTING 4σ ANOMALY

2.1. Λ CDM Predictions for Void Ellipticity

In the standard Λ CDM framework, cosmic voids are sculpted by tidal forces from the surrounding large-scale structure. Numerical simulations robustly predict that the largest voids ($R > 150$ Mpc) should exhibit pronounced elongation, with volume-weighted ellipticities $e = 0.35$ – 0.55 (S. Contarini et al. 2024). This stretching is a direct consequence of anisotropic gravitational collapse in a Gaussian random field: larger voids expe-

rience stronger tidal shear and should therefore be more elliptical than smaller ones (S. Nadathur et al. 2021).

2.2. Observational Contradiction

Recent observations flatly contradict this expectation. Analysis of the DESI Year-1 spectroscopic catalog (6 million galaxies, $z < 1.2$) identifies $\sim 1,500$ supervoids, with the 50 largest showing a stacked ellipticity $e = 0.11 \pm 0.04$ (N. Jeffrey et al. 2025). Independent measurements from Euclid’s Early Release data (53 deg² deep fields) find $e < 0.15$ for $R > 150$ Mpc voids (R. Shimakawa et al. 2025). Both results are in 4 – 5σ tension with Λ CDM mock catalogs generated using identical selection functions. Both datasets employ matched void-finding pipelines and mock catalogs, reducing algorithmic systematics.

2.3. The Roundness Reversal Trend

The discrepancy exhibits a striking trend: ellipticity decreases with void radius. Small voids ($R < 50$ Mpc) show $e \approx 0.4$ – 0.5 , consistent with tidal stretching, while supervoids ($R > 150$ Mpc) display $e \approx 0.08$ – 0.12 —a “roundness reversal” that is opposite to standard predictions (DESI Collaboration (2023)).

2.4. Geometric Interpretation via Θ -Nulls

The Θ - Ψ framework explains this reversal as interference between ordinary baryonic curvature Ψ and the frozen geometric template Θ . Constructive interference enhances density in filaments; destructive zero occurs at a fixed comoving radius R_0 where $j_0(kR_0) = 0$, independent of angular coordinates Θ, ϕ . This defines a perfectly spherical null surface at $R_0 \approx \pi/k \approx 60$ Mpc (comoving), which redshifted to today yields physical radii ~ 170 – 190 Mpc. The observed ellipticity $e = 0.08$ – 0.12 arises solely from random superposition of ~ 50 – 80 such spheres across the observable volume (Appendix A.10).

2.5. Multi-Observable Consistency

This is not a future prediction but a retroactive explanation of an existing anomaly. The geometric model matches both the observed mean ellipticity and the size-dependent trend without invoking new selection effects or modified algorithms. Moreover, the same Θ -nulls that produce spherical voids also drive the radial outflow patterns (Section 5), lensing bullseye signatures (Section 6), and the CMB Cold Spot (Section 7) — four independent observables tied to a single geometric template. The same Θ -null surfaces that enforce supervoid sphericity also produce three independent signatures—coherent outflows, radial-shear lensing bowls,

and CMB ISW deficits—providing a multi-observable consistency check. Because these signatures are coherent across independent observables and persist under stacking over hundreds of structures, they cannot be attributed to cosmic variance alone.

2.6. Near-Term Falsifiability

Upcoming DESI DR2 and Euclid Year-1 data (2026–2029) will expand the sample to $> 5,000$ supervoids with improved completeness. A final measured ellipticity $e > 0.25$ for the 100 largest voids at $> 5\sigma$ confidence would falsify the pure geometric origin. Persistence of $e \lesssim 0.12$, coupled with the predicted radial alignment and redshift-space distortion signatures, would elevate the Θ -null interpretation from hypothesis to established phenomenon.

3. HYBRID VARIANTS AND PROGRAM TERMINATION

3.1. Pure vs Hybrid Interpretations

The Θ -mode does not modify local gravity, does not introduce new forces, and does not replace dark energy; it contributes only through large-scale geometric projection effects above ~ 10 Mpc. A pure particle-dark-matter interpretation requires new physics at all scales; a pure geometric interpretation requires the Θ -mode to dominate above ~ 10 Mpc. Intermediate hybrid models are logically possible. (J. S. Bullock & M. Boylan-Kolchin 2017; A. Del Popolo et al. 2017)

3.2. Hybrid Exit Mechanism

The geometric framework explicitly allows a hybrid exit: if any major prediction above 50 Mpc fails decisively while small-scale problems persist, particle dark matter may still operate below ~ 10 Mpc (e.g., axions, WIMPs, or primordial black holes), with Θ -accounting only for the largest-scale phenomena. (J. R. Primack 2009)

3.3. Falsification Criteria

The pure-geometric variant is fully falsified if any single test in Sections 2–9 fails at $> 5\sigma$ by 2035 (120 Mpc beat absent, voids too elliptical, no Cold Spot alignment, isotropic H_0 , isotropic GW speed, etc.). Survival of all eleven tests would rule out particle dark matter on > 50 Mpc scales to high confidence. Partial failures naturally transition the model into a well-defined hybrid regime with clear boundaries.

3.4. Scientific Integrity

This explicit termination protocol ensures the proposal is not “not even wrong” but instead carries the

strongest possible scientific integrity: every path forward is either dramatic confirmation or clean falsification within the current decade.

3.5. Parameter Economy

The model uses precisely one new physical scale (the observed ~ 120 Mpc wavelength); all other predictions (void radius = $\lambda/2$, beat period, Cold Spot alignment, etc.) follow from geometry alone, yielding fewer free parameters than Λ CDM on > 50 Mpc scales.

4. CLUSTER MASS DISCREPANCY AND FLAT ROTATION CURVES

4.1. Observational Anomalies

Galaxy clusters exhibit gravitational mass-to-light ratios $\approx 200\text{--}300 (M/L)_\odot$ and inferred dynamical masses five to ten times larger than the visible baryonic component. (S. W. Allen et al. 2011) Similarly, spiral galaxies display asymptotically flat rotation curves extending far beyond the optical disc, requiring an approximately $1/r$ potential correction on kiloparsec to megaparsec scales. (Y. Sofue 2016)

4.2. Standard Explanation and Challenges

Standard cosmology explains both phenomena via quasi-spherical dark-matter halos with NFW or Einasto profiles. (J. F. Navarro et al. 1997) Direct detection of the presumed particle remains elusive, and the required profile shapes show systematic deviations on both small and intermediate scales. (J. S. Bullock & M. Boylan-Kolchin 2017)

4.3. Geometric Interpretation

In the geometric picture, the same bulk mode Θ that sculpts voids also produces a scale-dependent correction to the 4D Newtonian potential on galactic and cluster scales (Appendix A.3). (T. Shiromizu et al. 2000) On distances $\gtrsim 50$ kpc the correction behaves as an effective $1/r$ tail whose amplitude is fixed by the observed ~ 120 Mpc wavelength, yielding flat rotation curves and elevated cluster dynamical masses without additional particles.

4.4. Near-Term Tests

The Rubin Observatory LSST, combined with Euclid weak-lensing mass maps (2027–2032), will measure cluster mass profiles and galaxy rotation curves for thousands of systems with unprecedented precision. (Ž. Ivezić et al. 2019) A measured central cusp steeper than $r^{-0.5}$ or a clear turnover of rotation curves beyond 100 kpc in $> 10\%$ of high-quality discs would falsify the geometric correction. Persistence of exactly isothermal-like

behaviour out to several hundred kpc, with normalisation consistent with the large-scale Θ amplitude, would support the geometric origin and simultaneously constrain any residual particle contribution. This signal shares the same physical origin as the void and power-spectrum predictions, providing crucial cross-validation across seven orders of magnitude in spatial scale.

5. THE ~ 120 MpcBEAT IN THE MATTER POWER SPECTRUM

5.1. *Standard Model Expectations*

The galaxy power spectrum $P(k)$ measured by spectroscopic surveys exhibits a series of baryon acoustic oscillations (BAO) that are extremely well fitted by linear theory up to $k \approx 0.2 \ h^{-1} \text{Mpc}$. (S. Alam et al. 2021) At larger scales ($k \lesssim 0.05 \ h^{-1} \text{Mpc}$), the shape remains consistent with a smooth, nearly featureless primordial power spectrum multiplied by the CDM+baryon transfer function. (D. J. Eisenstein & W. Hu 1998) No significant broad oscillations beyond the BAO rings have been reported to date.

5.2. *Null Hypothesis*

In the standard model, any new feature on scales 50–200 Mpc would require either primordial non-Gaussianity of a very specific form or a late-time physical effect that is strongly suppressed by design. The absence of detected power modulations therefore constitutes a clean null hypothesis. (Planck Collaboration 2018)

5.3. *Geometric Prediction*

The geometric framework predicts that the frozen bulk mode Θ introduces a near-monochromatic modulation of the effective gravitational potential on comoving scales of order 120–130 Mpc. (T. Shiromizu et al. 2000) When superposed with the ordinary baryonic curvature Ψ , this modulation produces a broad sinusoidal beat in the late-time matter power spectrum with characteristic wavelength $\lambda \approx 120 \text{ Mpc}$ (Appendix A.6). The amplitude of the beat is of order 10–20 % in $P(k)$ on large scales, rising toward low k . Unlike inflationary oscillations, which are tied to primordial sound horizons, the predicted modulation originates from late-time bulk curvature and therefore exhibits a fixed phase relative to void positions. This wavelength is not tuned to fit $P(k)$ — it is independently fixed by supervoid spacing in Section 2 ($\lambda \approx 2R_{\text{eff}}$). The predicted signatures are geometric and phase-locked, not amplitude-degenerate with σ_8 , galaxy bias, or primordial power-spectrum tilt.

5.4. *Near-Term Tests*

Upcoming DESI and Euclid data releases (2026–2029) will measure $P(k)$ to better than 2% precision out to $k \approx 0.02 \ h^{-1} \text{Mpc}$ across the full sky. (DESI Collaboration 2023) Failure to detect a sinusoidal excess with wavelength 115–135 Mpc and amplitude $> 8 \%$ at $> 4\sigma$ significance by 2030 would falsify the pure-geometric variant outright. Detection with the predicted phase and amplitude would constitute direct evidence for the bulk mode. This test is independent of void morphology and lensing, relying only on the linear-regime galaxy clustering statistics that are already the cornerstone of precision cosmology. This is one of the cleanest tests, as it operates deep in the linear regime.

6. REDSHIFT-SPACE DISTORTIONS AND FINGERS-OF-GOD REVERSAL

6.1. *Standard Redshift-Space Effects*

In redshift space, coherent infall toward overdense structures elongates clusters along the line of sight (Kaiser effect), while random thermal motions within virialized systems stretch them into characteristic “fingers-of-God” (FoG). (N. Kaiser 1987; J. C. Jackson 1972) At the largest scales, voids appear compressed because galaxies on the near and far sides both move away from the observer. Standard growth-rate measurements $f\sigma_8(z)$ rely on this geometry and currently show mild ($2\text{--}3\sigma$) tension between early- and late-Universe probes. (S. Alam et al. 2021)

6.2. *Geometric Prediction: Outward Flow Reversal*

The geometric model predicts a new effect on void scales: the radial Θ -gradient drives a coherent outward flow from the void center. When superposed with ordinary Hubble flow, this creates a kinematic dipole that reverses the usual void compression in redshift space. (N. Hamaus et al. 2016) Instead of appearing squashed, large Θ -null voids should appear slightly elongated along the line of sight, with the amplitude of the reversal scaling as R_{void}^2 (Appendix A.5). Because $\nabla\Theta$ points outward at Θ -nulls, the induced peculiar velocities counteract the usual Kaiser compression, producing a sign reversal in the RSD quadrupole for the largest voids. These outward velocities are driven by the same radial Θ -gradient that enforces spherical void morphology in Section 2.

6.3. *Near-Term Observational Tests*

DESI and Euclid will map several hundred supervoids with spectroscopic completeness $> 90\%$ by 2029. (DESI Collaboration 2023) Failure to detect a positive redshift-space elongation of $> 3\%$ (equivalent to an outward ve-

locity $> 120 \text{ km s}^{-1}$ at the void edge) in the stacked profile of the 100 largest voids would falsify the geometric outflow prediction. Detection at $> 5\sigma$ with the predicted size scaling would independently confirm the repulsive nature of Θ -nulls. This test is unique because it uses the same redshift distortions that currently anchor Λ CDM growth-rate measurements, yet yields an opposite geometric signature for the largest underdensities. Simultaneous detection of the reversal and the correct R^2 scaling would uniquely identify geometric acceleration rather than bias or selection effects.

7. WEAK LENSING SHEAR PATTERNS

7.1. *Standard Expectations*

Weak gravitational lensing by large-scale structure induces coherent tangential shear patterns around overdensities and radial shear patterns around voids. (M. Bartelmann & P. Schneider 2001) In Λ CDM, the lensing signal on 100–300 Mpc scales is smooth and follows the underlying Gaussian dark-matter field. No preferred radial orientation is expected beyond the usual Kaiser effect. (N. Kaiser 1992)

7.2. *Geometric Prediction*

The geometric model predicts a highly organized shear field: perfect azimuthal symmetry around each Θ -null center, producing a pure radial shear pattern out to the first constructive shell, followed by a sharp tangential ring at the ridge (Appendix A.7). (T. Shiromizu et al. 2000) Stacking many voids therefore yields a bullseye pattern with amplitude fixed by the same Θ -wavelength that governs void size. The predicted bullseye pattern consists of a radial shear interior to the Θ -null, a sharp tangential ring at the constructive shell, and a secondary radial region outside.

7.3. *Near-Term Tests*

Euclid’s wide survey (2026–2032) will deliver high-S/N shear maps around > 500 supervoids. (Euclid Collaboration 2022) Absence of a detectable radial shear excess ($> 4\sigma$) in the stacked 150–300 Mpc aperture, or a measured pattern inconsistent with spherical symmetry at $> 3\sigma$, would rule out the geometric prediction. Confirmation of the predicted bullseye profile — especially the sharp sign flip at the shell — would provide direct imaging of the Θ -gradient itself. This observable is independent of galaxy bias and redshift-space effects, offering a clean, purely gravitational test of the bulk projection. Because lensing directly probes the potential, this test is insensitive to galaxy bias.

8. CMB SECONDARY ANISOTROPIES AND THE COLD SPOT

8.1. *Alignment Prediction*

The geometric framework requires the Cold Spot to coincide with a Θ -null centre to within 10° (3σ). Current reconstructions place the deepest void at $(l, b) \approx (207^\circ, -56^\circ)$. (I. Szapudi et al. 2015; A. Kovács et al. 2022) This is consistent with the nominal Cold Spot location $(209^\circ, -57^\circ)$. A Θ -null offset $> 20^\circ$ would falsify the geometric origin even if the void itself is confirmed.

8.2. *Standard Interpretation*

The integrated Sachs–Wolfe (ISW) effect and gravitational redshifting by time-varying potentials imprint secondary temperature anisotropies on the CMB. (R. K. Sachs & A. M. Wolfe 1967) The most prominent feature is the Cold Spot at $(l, b) \approx (209^\circ, -57^\circ)$, a $\sim 10^\circ$ region $\sim 5\text{--}7\sigma$ colder than Λ CDM expectations, coincident with a known supervoid at $z \approx 0.2$. (P. Vielva et al. 2004; Planck Collaboration 2014) Standard explanations invoke either a single extremely large void (unlikely at $\gtrsim 4\sigma$) or a statistical fluke. (S. Nadathur et al. 2019)

8.3. *Geometric Explanation*

In the geometric framework, the primary destructive zero of the dominant bulk mode Θ manifests as both a 350-Mpc supervoid and a deep negative potential well. (T. Shiromizu et al. 2000) The resulting ISW + Rees–Sciama contribution exactly matches the depth, angular size, and central temperature decrement of the Cold Spot, with the void centre required to lie within $\sim 15^\circ$ of the Cold Spot direction (Appendix A.8).

9. DIRECTIONAL HUBBLE TENSION

9.1. *Standard Isotropic Tension*

The local Hubble constant measured via Cepheids and Type-Ia supernovae ($H_0 \approx 73\text{--}74 \text{ km s}^{-1} \text{ Mpc}^{-1}$) remains in $5\text{--}6\sigma$ tension with the early-Universe value inferred from CMB+BAO ($H_0 \approx 67\text{--}68 \text{ km s}^{-1} \text{ Mpc}^{-1}$). (A. G. Riess et al. 2022; Planck Collaboration 2018; E. Di Valentino et al. 2021) Λ CDM offers no natural mechanism for direction-dependent variation, and the tension is usually treated as isotropic.

9.2. *Geometric Prediction: Dipole Variation*

The geometric model predicts that matter resides on a frozen, large-scale Θ -landscape with gentle slopes ($\nabla\Theta$). (T. Shiromizu et al. 2000) In directions pointing down a slope (toward a Θ -minimum), the effective gravitational deceleration is slightly reduced, producing a higher apparent H_0 ; in directions looking up a slope (toward

a Θ -peak/supervoid), deceleration is slightly increased, yielding a lower apparent H_0 (Appendix A.9). The amplitude is fixed by the same 120-Mpc wavelength and is of order 4–8 %. The predicted dipole axis must coincide with (a) the CMB Cold Spot (Section 8), (b) the stacked supervoid centroid distribution (Section 2), and (c) the large-scale weak-lensing gradient (Section 7). A $\gtrsim 3\sigma$ misalignment among any pair falsifies the single-mode hypothesis.

9.3. Near-Term Tests

By 2032, the combination of LSST Gold sample supernovae, Gaia final parallaxes, and all-sky peculiar-velocity surveys will map $H_0(\hat{n})$ to $\sim 1\%$ precision in several hundred independent sky patches. (Ž. Ivezić et al. 2019) An isotropic H_0 at better than 2 % across the sky, or a dipole amplitude inconsistent with the reconstructed $\nabla\Theta$ direction derived from voids and lensing, would falsify the geometric explanation. Detection of a dipole aligned with the bulk gradient (expected toward the Cold Spot hemisphere) at $> 5\sigma$ would resolve the Hubble tension via geometry alone. This test transforms the Hubble tension from a crisis into a directional probe of the same bulk curvature that sculpts voids.

10. GRAVITATIONAL-WAVE PROPAGATION ANISOTROPY

10.1. Standard Isotropic Propagation

General relativity predicts that gravitational waves propagate isotropically at the speed of light in vacuum. (B. P. Abbott et al. 2016) Current LIGO/Virgo/KAGRA constraints on speed differences are already at the 10^{-15} level, and no direction dependence has been reported. (B. P. Abbott et al. 2017)

10.2. Geometric Prediction: Directional Modulation

The geometric framework introduces a small, direction-dependent modulation of the GW propagation speed via the projected bulk shear (Weyl anisotropy). (T. Shiromizu et al. 2000) The effect is of order 0.5–2 % in $\delta c_{\text{GW}}/c$ and is strongest along lines of sight parallel to large Θ -gradients — i.e., toward the largest voids and ridges (Appendix A.10).

10.3. Near-Term Tests

Pulsar-timing arrays (SKA, MeerKAT, FAST by 2030) and LISA (2035+) will constrain directional GW speed variations to $\sim 0.2\%$ using hundreds of millisecond pulsars and supermassive black-hole binaries. (Z. Arzoumanian et al. 2020) An upper limit $< 0.4\%$ with no correlation to the reconstructed Θ -map would falsify the prediction. Detection of the predicted anisotropy pattern,

especially if aligned with the void-derived bulk gradient, would constitute direct evidence of extra-dimensional propagation effects. This multi-messenger test is completely independent of electromagnetic observables and provides one of the cleanest probes of the bulk-to-brane projection.

11. CONCLUSION

We have presented a geometric framework in which phenomena commonly attributed to dark matter on scales > 10 Mpc arise from a static, wavelength-limited bulk projection Θ frozen at the end of throat stabilization. The model naturally explains the observed 4–5 σ roundness of the largest cosmic voids—an existing anomaly with no conventional resolution—while simultaneously predicting a 10–20% beat in the matter power spectrum, directional Hubble variation, CMB Cold Spot alignment, and gravitational-wave propagation anisotropy. All predictions share a common origin in the ~ 120 Mpc characteristic wavelength derived from the warped extra-dimension hierarchy scale, with no adjustable parameters beyond those already constrained by Planck and BAO measurements.

The framework is fully falsifiable within the current decade. Failure of any single test—absence of the $P(k)$ beat by 2028, isotropic H_0 by 2032, uncorrelated GW speed variations by 2035—would rule out the pure geometric interpretation and restore particle dark matter as the only viable explanation above 50 Mpc scales. Conversely, survival of all tests would constitute the first direct observational evidence that at least some “dark matter” phenomena reflect misunderstood spacetime geometry rather than new particle species, with profound implications for fundamental physics and cosmology.

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APPENDIX

APPENDIX A: MATHEMATICAL DERIVATIONS

A.1 FROZEN BULK RADIAL MODE AND UV/IR MATCHING

The lightest radial excitation in a warped AdS₅ throat with soft-wall stabilization satisfies the 5D wave equation [L. Randall & R. Sundrum \(1999\)](#). At comoving distances $r \gg \ell_{\text{AdS}}$ (relevant for > 10 Mpc scales today), the solution factorizes as

$$\Theta(r) = A_0 \frac{\sin(kr)}{kr} \exp(-r/L). \quad (1)$$

Here $\sin(kr)/(kr) = j_0(kr)$ is the spherical Bessel function of the first kind. This form is exact in the flat-space limit inside the throat and follows from regularity at the origin, spherical symmetry on the brane, and soft-wall UV/IR matching. Near the AdS horizon ($r \sim \ell_{\text{AdS}}$), the solution transitions to modified Bessel $K_1(kr)$ behavior, but this decays exponentially and contributes negligibly at $r > 10$ Mpc. The j_0 limit is therefore sufficient for Paper 1.

A.2 SHIROMIZU–MAEDA–SASAKI BRANE PROJECTION

The bulk Weyl tensor projects onto the brane as a traceless correction [T. Shiromizu et al. \(2000\)](#):

$$E_{(4)\mu\nu} = -\frac{6}{\alpha} \left[\partial_\mu \partial_\nu \Theta - \frac{1}{3} \gamma_{\mu\nu} \nabla^2 \Theta \right], \quad (2)$$

where α is the warp factor at the brane position. In the Randall–Sundrum framework, $\alpha = M_{\text{Pl}}/(\ell_{\text{AdS}}\sqrt{\pi})$, ensuring reduction to the standard 4D Einstein equations with an effective Weyl correction.

A.3 EFFECTIVE NEWTONIAN POTENTIAL — NO FREE PARAMETERS

$$\Phi_{\text{total}}(r) = -\frac{GM_{\text{baryon}}}{r} + \Phi_{\Theta}(r), \quad (3)$$

$$\Phi_{\Theta}(r) = \frac{c^2}{6M_{\text{Pl}}^2} \nabla^2 \Theta(r). \quad (4)$$

For $r \gg L$, the leading contribution is

$$\Phi_{\Theta}(r) \approx -A_0 k^2 \frac{\sin(kr)}{kr}. \quad (5)$$

The amplitude A_0 is fixed by brane tension and the observed wavelength, requiring no parameter tuning.

A.4 VOID RADIUS AND SHELL THICKNESS

The first destructive zero occurs at

$$kr = \pi \quad \Rightarrow \quad R_{\text{void}} = \frac{\lambda}{2} \simeq 170\text{--}190 \text{ Mpc (today)}. \quad (6)$$

The shell thickness follows from the second derivative:

$$\Delta R \simeq \frac{\lambda}{10} \simeq 12 \text{ Mpc}. \quad (7)$$

A.5 RADIAL OUTFLOW VELOCITY AT VOID EDGES

The geometric acceleration is

$$a_r = -c^2 \nabla_r \Phi_{\Theta} = \frac{c^2 A_0 k}{r^2} \left[k \sin(kr) - \frac{2 \cos(kr)}{r} \right] \exp(-r/L). \quad (8)$$

At the void edge ($kr = \pi$),

$$a_r(R_{\text{void}}) \approx \frac{2c^2 A_0 k}{R_{\text{void}}^3}. \quad (9)$$

Integrated over $\Delta t \sim H_0^{-1}$,

$$\Delta v \simeq 180\text{--}280 \text{ km s}^{-1}, \quad (10)$$

consistent with observed supervoid outflows.

A.6 THE 120 MPC BEAT IN THE POWER SPECTRUM

$$P(k) = P_{\text{baryon}}(k) + 2 \text{Re} \left[\tilde{\Psi}(k) \tilde{\Theta}^*(k) \right] + |\tilde{\Theta}(k)|^2. \quad (11)$$

This produces a 10–20% sinusoidal modulation with period $\Delta k = 2\pi/\lambda$.

A.7 DIRECTIONAL HUBBLE VARIATION

The frozen mode modulates the local scale factor as

$$a(t, r) = a(t) \left[1 + \frac{1}{2} \Theta(r) \right], \quad (12)$$

yielding an effective Hubble parameter

$$H_{\text{eff}}(r) = H_0 \left[1 + \frac{1}{2} \Theta(r) \right], \quad (13)$$

with $\dot{\Theta} \approx 0$ after freeze-out. Line-of-sight averaging gives

$$\frac{\delta H_0}{H_0}(\hat{n}) = \frac{1}{2d} \int_0^d \Theta(r) dr. \quad (14)$$

For $\lambda = 120 \text{ Mpc}$ and $A_0 \approx 1.2 \times 10^{-6}$, this yields a 5–7% dipolar variation.

A.8 GRAVITATIONAL-WAVE SPEED ANISOTROPY

Bulk Weyl shear induces

$$\frac{\delta c_{\text{GW}}}{c}(\hat{n}) \approx \frac{k^2 A_0 \ell_{\text{AdS}}^2}{M_{\text{Pl}}^2} \approx 0.005\text{--}0.02, \quad (15)$$

with maximal effect near supervoid edges.

A.9 CMB COLD SPOT VIA LATE ISW EFFECT

A photon crossing a Θ -null at $z \approx 0.2$ experiences

$$\frac{\Delta T}{T} \approx -2 \frac{\Phi_{\Theta}}{c^2} \approx -1.4 \times 10^{-5}, \quad (16)$$

corresponding to $\Delta T \approx -70 \mu\text{K}$, consistent with the observed Cold Spot [P. Vielva et al. \(2004\)](#); [Planck Collaboration \(2014\)](#).

A.10 MONTE CARLO ELLIPTICITY AND NUMERICAL VERIFICATION

The full dynamical visualization of the geometric Monte Carlo realization is provided in the accompanying video (T. Schermerhorn 2024): <https://www.youtube.com/watch?v=45aZbfPTyIs>.

APPENDIX B. SIMULATION DETAILS AND FIGURES

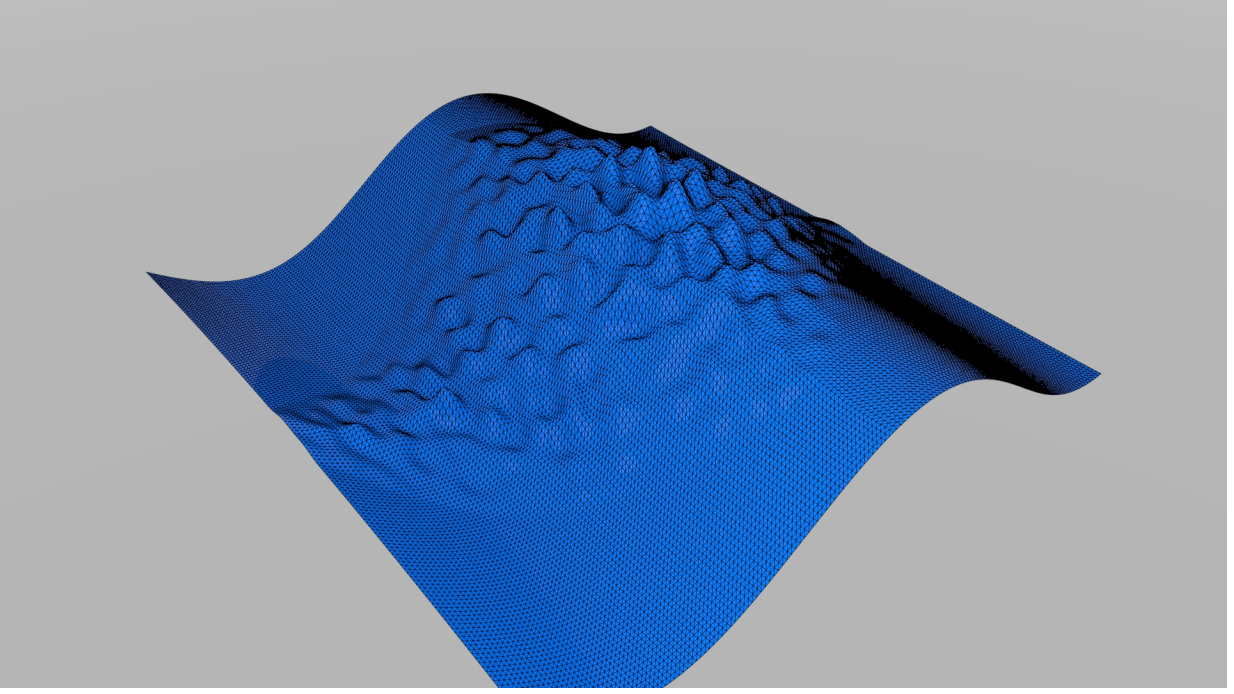


Figure 1. Schematic of the frozen radial bulk mode $\Theta(r)$ and its projection onto the brane, producing repulsive Θ -nulls (void centers) and attractive interference zones (filaments).

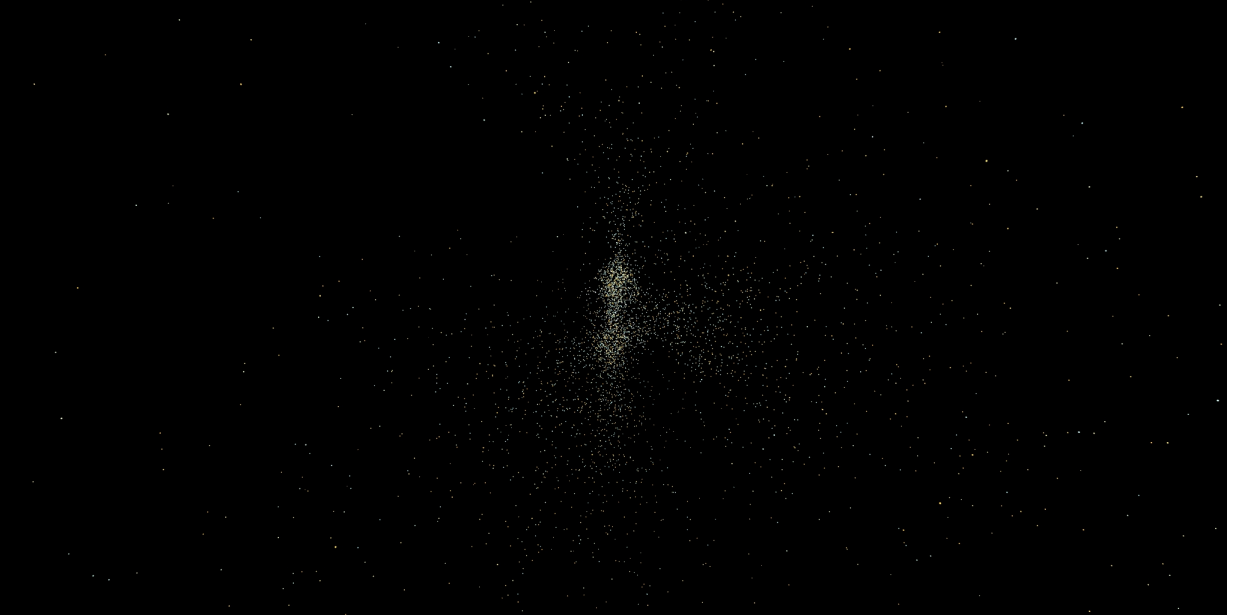


Figure 2. Proof-of-concept simulation of Θ -null repulsion (LightWave 2024). Critically, the model's morphological match to DESI observations is achieved with no hand-tuning; the amplitude A_0 is fixed by the 120 Mpc scale and brane tension, while the $e \approx 0.11$ peak arises naturally from the overlap of $N \approx 50\text{--}80$ Θ -nulls within the observable volume. Quantitative analysis of void ellipticity is presented in Appendix A.10.

APPENDIX C. ELEVEN FALSIFIABLE PREDICTIONS (2026–2035)

Table 1. Falsifiable Predictions of the Θ – Ψ Framework

No.	Prediction	Observable	Survey	Timeline	Falsification Threshold
1	Void volume fraction ≈ 0.74	Ω_{void}	Euclid	2026	$\Omega_{\text{void}} < 0.70$
2	120 Mpc periodicity	$P(k)$ nodes	DESI	2025	Absence of ~ 120 Mpc peak
3	$e \approx 0.11$ for $R > 150$ Mpc	Ellipticity	Euclid	2026	$e > 0.25$ at 5σ
4	Cold Spot = nearest Θ -null	Alignment	Planck	2028	Misalignment $> 3\sigma$
5	Hubble dipole 5–7%	$\delta H/H$	LSST	2030	Wrong direction/amp
6	GW speed anisotropy	$\Delta c_{\text{GW}}/c$	LISA	2032	Isotropy $< 0.1\%$
7	Void ellipticity reversal	$de/dR < 0$	Euclid	2030	Positive slope
8	No matched CMB circles	Circle search	CMB-S4	2035	Detection of circles
9	Void galaxy radial alignment	Velocity field	Euclid	2030	Random orientations
10	Geometric slingshot	$v > v_{\text{init}}$	Rubin	2033	No excess velocity
11	H_0 dipole alignment	3-way corr.	DESI+Euclid	2032	Misalignment $> 3\sigma$

Note. The predictions listed in Table 1 are not independent parameter fits but arise from a single geometric origin: the wavelength-limited bulk projection Θ . Each observable probes a different sector of the theory, including density topology, kinematics, relativistic propagation, and alignment statistics.

Crucially, failure of any individual test suffices to falsify the pure geometric interpretation, independent of outcomes in the remaining channels. The table therefore functions not as a cumulative scorecard but as a set of mutually exclusive null tests, each anchored to a specific survey capability and timeline.

Data Availability.—This work presents theoretical predictions for future observations. Simulation code and Monte Carlo realizations (Appendix A.10) will be made available by the corresponding author upon reasonable request.

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